

bloodstream isolates between 2007 and 2010 in Phnom Penh found a high prevalence of ESBL producers, but no CPE.¹⁰

In conclusion, we report two temporally and geographically linked patients with community-onset urinary tract infection caused by the same OXA-48-producing *E. coli* clone. The association between OXA-48 and Cambodia in NZ suggests the Cambodian patient may have acquired the organism during recent travel and indirectly transmitted to the second patient via unknown community transmission pathways. It is possible that OXA-48-positive CPE are under-recognized in NZ due to the challenges of laboratory detection. Screening with adequate selective media for CPE is highly recommended. The forthcoming national response plan to CPE in NZ has potential to help address some of these issues.

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Transparency declarations

None to declare.

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Innovations for the treatment of a complex bone and joint infection due to XDR *Pseudomonas aeruginosa* including local application of a selected cocktail of bacteriophages

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Sir,

A man in his early 60s had received chemotherapy (cisplatin/pemetrexed) with bevacizumab to treat disseminated non-small cell lung

cancer. External beam radiotherapy followed by cementoplasty was also performed for bone metastases located on the spine and on the right sacro-iliac joint. Two months later, a fistula occurred with clinical evidence of infection of the cement located in the right sacro-iliac joint (Figure S1A, available as [Supplementary data](#) at JAC Online). A surgery was required to remove the cement and to debride an abscess located in the psoas muscle. Negative-pressure wound therapy was implemented by using a vacuum-assisted closure system (VAC® Therapy, KCI, USA). MSSA grew in culture and the patient received intravenous cefazolin, replaced by imipenem/cilastatin as he developed a catheter-related bacteraemia due to ceftazidime-resistant *Pseudomonas aeruginosa*. Despite the therapy, the patient still had fever with purulent local secretion without healing, and a CT scan revealed persistent osteomyelitis (Figure S1B–D). A surgical bone biopsy revealed infection with an XDR *P. aeruginosa* that was only susceptible to polymyxins (colistimethate MIC <0.5 mg/L) and ceftolozane/tazobactam (MIC = 4 mg/L), a recently approved antibiotic for the treatment of complicated urinary tract infections and complicated intra-abdominal infections. In the hope of being able to use as quickly as possible immune checkpoint inhibitors for the non-small cell lung cancer, we proposed a two-stage surgical approach with surgical debridement and then surgical reconstruction with a local muscle and skin and soft tissue flap. We also combined this approach with several innovations to minimize healing time: local application of a selected cocktail of bacteriophages for compassionate use, local colistimethate therapy and off-label use of ceftolozane/tazobactam. Under the supervision of the French National Agency for Medicines and Health Products Safety (ANSM), bacteriophages were selected among the anti-*P. aeruginosa* bacteriophage library of Pherecydes Pharma (Romainville, France). The selection was based on a susceptibility test (which we propose to name ‘phagogram’) performed on the patient’s strain (Figure 1). The four most active bacteriophages were chosen and then manufactured by Pherecydes, outside of the Good Manufacturing Practices (GMP) structure but in accordance with GMP guidelines and with the agreement of the ANSM. The cocktail was assembled by the hospital pharmacist as a magistral preparation under laminar air flow and diluted in saline solution (0.9% NaCl; final volume 30 mL; final concentration of each bacteriophage 1.2–9.7 × 10⁸ cfu/mL). During the surgery, a debridement of the necrotic bone was performed, and 10 mL of the cocktail was brought into contact with the bone in the cavity (Figure S1E). Then compresses suffused with 20 mL of the bacteriophage solution were placed in the cavity before application of a waterproof dressing. The patient remained in ventral decubitus for 4 h to ensure that the phages remained in contact with the infected bone. As the patient had mild kidney injury, and as colistimethate is nephrotoxic, we decided to use local administration of this antibiotic. A VACULTA® Therapy System (KCI, USA) was used to administer locally 36 million international units (MIU)/day of colistimethate (Colimycine® Sanofi-Aventis, France), corresponding to 1200 mg of colistin base activity. Twelve cycles/day were scheduled as follows: 5 min of instillation, 20 min break and 95 min of negative pressure. In addition, ceftolozane/tazobactam was given intravenously (1 g/500 mg/8 h). Vacuum-assisted closure therapy was stopped for 4 h to allow local phage administration, which was performed in the same manner (with suffused compresses) every 3 days, totalling four administrations. At the time of surgical reconstruction at day 14, the macroscopic aspect was extremely favourable, and muscle and skin and

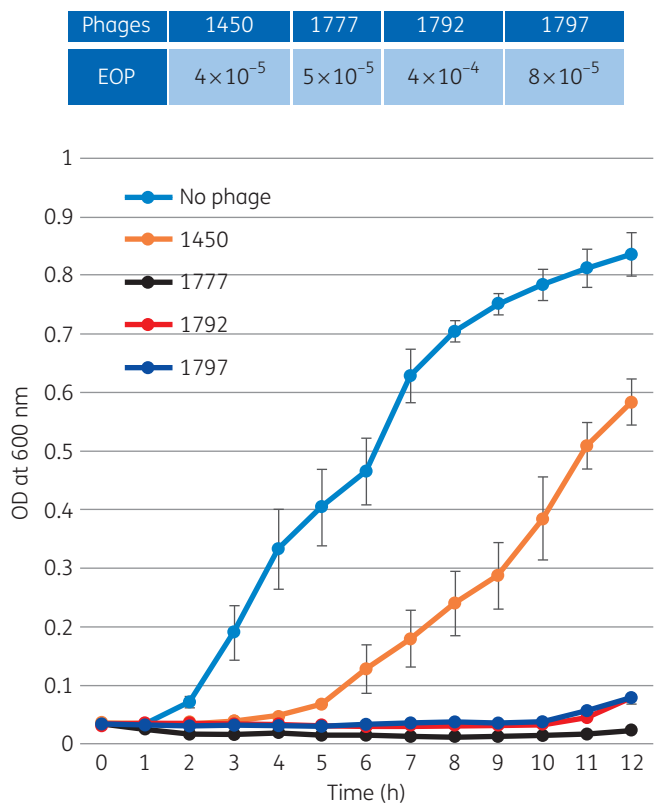


Figure 1. Ten anti-*P. aeruginosa* natural strictly lytic phages were tested [efficiency of plating (EOP) analysis and growth inhibition in liquid culture]. The EOP score is defined by the phage titre on the patient’s strain divided by the phage titre on a reference strain showing the maximum titre observed.⁸ The closer the score is to 1, the more effective the phage is, indicating that it will be active at low concentration. For the measurement of growth inhibition in liquid medium, the bacterium was cultured in a 96-well plate at a starting concentration of 10⁸ cfu/mL, with or without phage at a concentration of ~10¹⁰ pfu/mL. The bacterial concentration was monitored over time by OD at 600 nm. In the absence of phage, we did not observe growth inhibition of the patient’s strain. When the 1777, 1792 and 1797 phages were added individually to the culture, we observed complete growth inhibition of the patient’s strain. When the patient’s strain was cultured with 1450 phage, no growth was detected during the first 5 h. After 6 h, we detected a recovery of growth that could correspond to bacterial resistance to phage 1450. These four phages were selected based on their activity and availability. They were characterized by deep sequencing ×5–10 000 [two phages belong to Myoviridae (65–66 kbp) and two phages belong to Podoviridae (44–45 kbp)]. An extensive quality control was performed for titre (10¹⁰ pfu/mL after finishing), identity and purity (RFLP), sterility, pH (7.1), endotoxin (<1–5 endotoxin units (EU)/mL for 10¹⁰ pfu/mL), host cell DNA (<limit of detection) and host cell protein (<20 µg/mL). This figure appears in colour in the online version of JAC and in black and white in the print version of JAC.

soft tissue flap reconstruction was performed (Figure S1F–H). No bacteria grew in the culture, and healing was rapid (Figure S1I). Unfortunately, the patient developed neurological impairment of both legs, due to significant progression of the spine metastases, leading to compression. As a consequence, immunotherapy was not performed and the patient died at day 45.

Emergence of XDR and pandrug-resistant pathogens worldwide revives the absolute need for alternatives to systemic antibiotics.¹ XDR *P. aeruginosa* is increasingly prevalent, especially in the ICU setting.² In patients infected with XDR *P. aeruginosa*, treatment options are often limited to last-resort antibiotics, such as ceftolozane/tazobactam and polymyxins.^{1,2} In patients with complex bone and joint infections, a personalized multidisciplinary approach is required to determine the optimal medico-surgical strategy. In the case presented here, the accumulation of medical and surgical complications pushed us to use innovative synergistic treatment combining off-label use of ceftolozane/tazobactam, local colistin therapy, and a cocktail of personalized bacteriophages.^{3–5} Bacteriophages were first described in 1917, and remained a popular treatment throughout the 20th century in Eastern Europe, especially for patients with osteomyelitis. By their nature, lytic bacteriophages are good candidates for antibacterial therapy. Indeed, in comparison with antibiotics, they target a specific bacterium, as long as the corresponding bacteria is present. This phenomenon is unique as it is exponential and in theory self-sustaining after a single or a few administrations.^{6,7} The lytic bacteriophages used here were selected for their high *in vitro* efficacy against the *P. aeruginosa* strain responsible for the infection, and were produced via a process in accordance to GMP guidelines that was implemented according to EMA standards. Performance of a 'phagogram' to select the most efficient cocktail of bacteriophages is probably a relevant approach for the optimal treatment of such XDR infections. Bacteriophage technology is particularly promising in patients with XDR bone and joint infection as: (i) bacteriophages and antibiotics are synergistic;^{4,5} (ii) there is no cross-resistance between antibiotic resistance and bacteriophage resistance;^{6,7} and (iii) some *in vitro* and animal models have demonstrated that bacteriophages could have antibiofilm activity.^{4,5}

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Transparency declarations

C. F., C. P. and G. L'hostis are employed by the commercial company Pherecydes Pharma. All other authors: none to declare.

Supplementary data

Figure S1 is available as [Supplementary data](#) at JAC Online.

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